

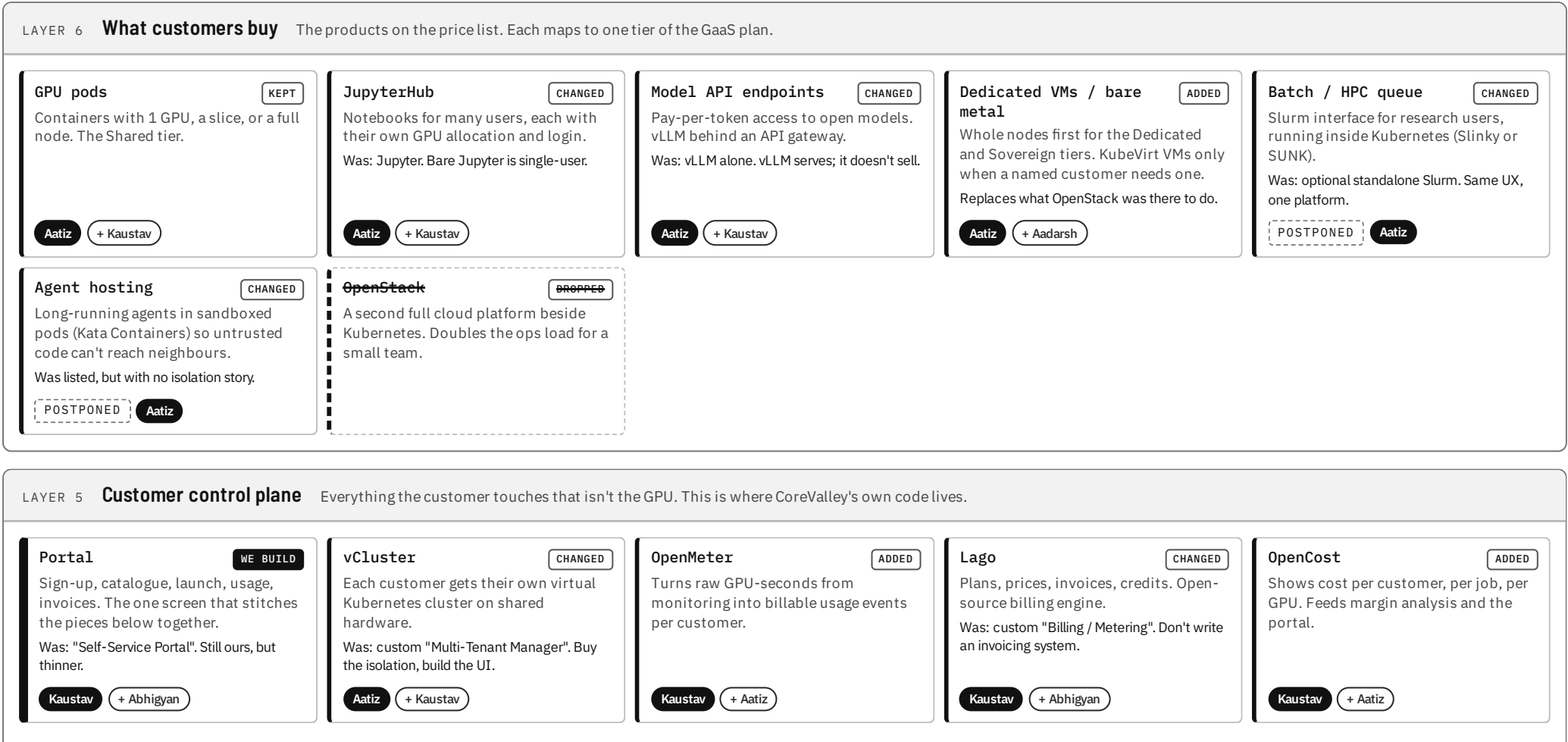
CoreValley Stack v2

The CoreValley platform stack, laid over the v1 draft so you can see what stays, what changes, what gets added, and the one thing that goes — now with one accountable owner on every tile.

KEPT in your draft, keep as is **CHANGED** in your draft, swapped or reshaped **ADDED** not in your draft **WE BUILD** the custom code CoreValley owns **DROPPED** in your draft, remove

POSTPONED owner named, work starts later

Aatiz CTO · infrastructure & platform **Aadarsh** network & security **Kaustav** fullstack · FDE **Abhigyan** MD · vendors, facility, commercial First pill = owner (accountable) · "+ name" = support



LiteLLM gateway

ADDED

API keys, rate limits, token counting, model routing in front of vLLM. Makes the model API a product.

Aatiz

+ Kaustav

Keycloak

KEPT

Single login for portal, notebooks, API and cluster. One realm per customer. Spelled "Keyclock" in v1.

Kaustav

+ Aadarsh

Status page + support desk

ADDED

Public uptime page and a ticket queue. Enterprise buyers ask for both before signing.

Kaustav

+ Abhigyan

Where it runs at the POC: on HPE's Kubernetes, hosted on the three DL325 management nodes, in CoreValley namespaces — pending HPE's written confirmation that our own charts and operators are allowed and survive their upgrades. All state on X10000 volumes with nightly Velero backups; everything deployed via Argo CD, so the move to the future phase is a sync, not a rebuild.

LAYER 4 **Platform control plane** Runs the cluster. Customers never see it; auditors and your engineers see all of it.

SCHEDULING & GPU

Kubernetes scheduler

KEPT

Decides which node a pod lands on.

Aatiz

Kueue

ADDED

Job queue: who waits, who gets priority, who is preemptible. The mechanism behind tiered pricing.

Aatiz

+ Kaustav

HAMi

KEPT

Splits one GPU between several pods (memory + compute slices). Shared tier.

Settled. First job: prove MIG mode on the H200s in the HPE box.

Aatiz

NVIDIA GPU Operator

KEPT

Installs drivers, CUDA runtime, DCGM and MIG config on every GPU node automatically.

Aatiz

NVIDIA Network Operator

ADDED

Exposes the fast GPU-to-GPU fabric (RDMA) to pods. Required for multi-node training.

Aatiz

+ Aadarsh

KubeVirt + Kata

ADDED

VMs inside Kubernetes (Dedicated tier) and hardened sandboxes (agents).

Aatiz

+ Aadarsh

DELIVERY & SUPPLY CHAIN

Argo CD (GitOps)

KEPT

Every cluster change is a git commit. Audit trail for free.

Aatiz

+ Kaustav

Harbor registry + pull-through cache

ADDED

Customers push images here; Docker Hub and NGC are mirrored locally so 20 GB images don't cross the border every job.

Aatiz

+ Aadarsh

Rancher

ADDED

Fleet view of all clusters (POC, future sites, customer vClusters). Ships with RKE2.

Aatiz

SECURITY & COMPLIANCE

Vault + External Secrets

ADDED

Secrets, API keys and certificates stored once, injected where needed.

Aadarsh

cert-manager

ADDED

Automatic TLS certificates for every endpoint.

Aadarsh

Kyverno

ADDED

Cluster rules: no privileged pods, signed images only, tenant quotas enforced.

Aadarsh

+ Aatiz

Falco

ADDED

Runtime alarm: flags a container doing something it shouldn't.

Aadarsh

Audit log shipping

ADDED

Kubernetes and Keycloak audit events written to immutable storage. SOC 2 asks for this first.

Aadarsh

Velero

ADDED

Backups of cluster state, restorable at the other site.

Aadarsh

+ Aatiz

NETWORK & CORE SERVICES

Cilium

ADDED

Pod networking with per-tenant firewall rules. Tenant A can't see tenant B's traffic.

Aadarsh

+ Aatiz

MetaLB + Envoy Gateway

ADDED

Load balancer and ingress on bare metal. How the internet reaches a customer's endpoint.

Aadarsh

+ Kaustav

DNS / NTP

KEPT

CoreDNS + external-dns; chrony against a local stratum-1 clock.

Aadarsh

LAYER 3 Observability Missing entirely from v1. Billing, SLAs and GPU health all depend on it.

DCGM exporter

ADDED

Per-GPU utilisation, memory, temperature, power, errors, every few seconds. The raw material for invoices.

Aatiz

+ Kaustav

Prometheus + Alertmanager

ADDED

Stores metrics, pages someone when a threshold is crossed.

Kaustav

+ Aatiz

Grafana

ADDED

Dashboards for ops, and a read-only view embedded in the customer portal.

Kaustav

+ Aatiz

Loki + OpenTelemetry

ADDED

Logs and traces, searchable per tenant.

Kaustav

+ Aadarsh

Facility telemetry

ADDED

Per-rack power, cooling and NEA feed data into the same Grafana. Your carbon-per-GPU-hour evidence.

Aadarsh

+ Abhigyan

LAYER 2 **Operating layer** What runs on each physical node.

RKE2

KEPT

Hardened Kubernetes distribution. Good fit for a sovereign, on-prem story.

Aatiz

Ubuntu + Talos

CHANGED

Ubuntu on the HPE POC (HPE dictates it); Talos Linux in the future phase where every node is identical.

Was: Ubuntu + Rocky. Rule: one OS per site, never two in one cluster.

Aatiz

+ Aadarsh

CUDA + drivers

KEPT

Delivered and versioned by the GPU Operator, not hand-installed.

Aatiz

MAAS / Tinkerbell

ADDED

Bare-metal provisioning: a new server gets its OS, firmware and join token without a human at the rack.

POSTPONED

Aatiz

+ Aadarsh

Node health loop

ADDED

node-problem-detector + DCGM diagnostics. A GPU throwing errors is drained automatically before a customer notices.

Aatiz

BMC / Redfish

ADDED

Out-of-band control: power-cycle and reimage a node remotely.

Aatiz

+ Aadarsh

LAYER 1 **Infrastructure** Left blank in v1. The two biggest capex lines after GPUs are in here.

Accelerators

ADDED

Current-gen NVIDIA only: H200, B200/B300, RTX PRO 6000 Blackwell. Mixed fleet, rack-level isolation option.

Abhigyan

+ Aatiz

Servers

ADDED

HPE for the POC (per quote); Dell PowerEdge or Supermicro rack-scale in the future phase.

Abhigyan

+ Aatiz

Storage · fast tier

ADDED

POC: HPE Alletra MP X10000 (file + S3) as shipped. Future phase: Ceph on our own nodes. Layer 5 only ever talks to "an S3 bucket" and "a volume", so the swap is invisible.

Aatiz

+ Abhigyan

Storage · object

ADDED

S3-compatible buckets: X10000 at the POC, Ceph RGW in the future phase. Every AI customer assumes this exists.

Aatiz

GPU fabric

ADDED

InfiniBand for the future phase (settled). POC uses HPE's 400GbE Spectrum fabric as shipped. Without a fabric, multi-node training doesn't scale.

Aadarsh

+ Aatiz

Front-end network + WAN

ADDED

100/400G Ethernet, out-of-band management network, upstream transit, and private links for Sovereign-tier customers.

Aadarsh

+ Abhigyan

Facility

ADDED

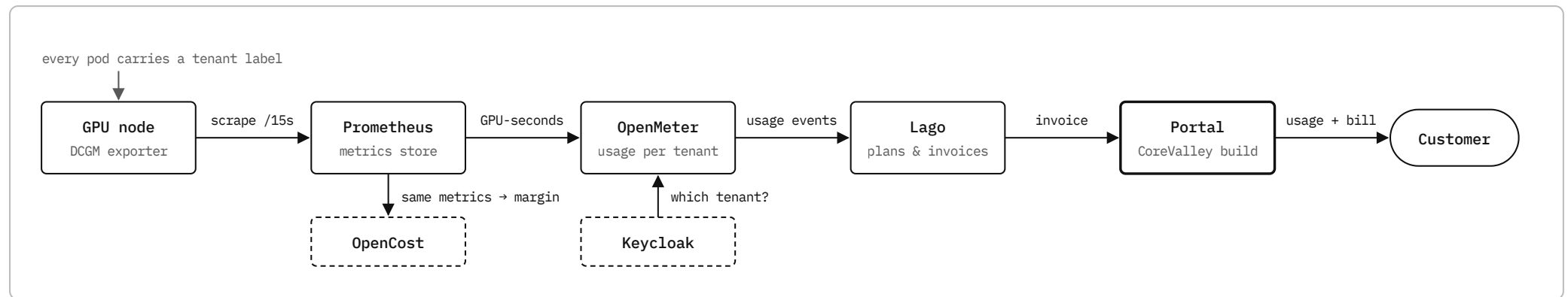
NEA dedicated feeder, UPS, cooling, DCIM + NetBox inventory. The compounding asset.

Abhigyan

+ Aadarsh

The mechanism v1 leaves out: how a GPU-second becomes an invoice

Your draft has a "Billing / Metering" box but nothing feeding it. In practice billing is a pipeline that starts at the GPU and ends at the customer's card, and every link is a separate tool. This is the single most important flow to get right during the POC, and it works the same whether the GPUs are HPE-managed or yours.



A GPU-second is measured on the node, stored in Prometheus, attributed to a tenant (via the pod label that Keycloak's login set), priced by Lago and shown in the portal. Remove any one box and you cannot bill by actual usage; you are back to flat monthly fees and arguments.

Side by side, layer by layer

LAYER	YOUR V1 DRAFT	PREFERRED V2	WHY THE DIFFERENCE MATTERS
Products	OpenStack, Jupyter, vLLM, GPU pods, agentic hosting	GPU pods, JupyterHub, model API (gateway + vLLM), VMs/bare metal (KubeVirt), Slurm-on-K8s, sandboxed agents	v1 mixes a platform (OpenStack) into the product list and has no way to sell tokens or dedicated nodes. v2 lists exactly what appears on the price list, one line per tier.
Customer control plane	Self-service portal: billing/metering, multi-tenant manager, service catalog (all custom)	Portal (custom) + vCluster + OpenMeter + Lago + OpenCost + LiteLLM + Keycloak + status/support	v1 writes three hard systems from scratch. v2 assembles them from proven parts and keeps the custom code to the portal, which is the only part customers judge you on.
Scheduling	K8s scheduler + HAMi	K8s scheduler + Kueue + HAMi	Without a queue there is no priority, no preemptible tier and no fair share between customers competing for the same GPUs.
Delivery	GitOps + Argo CD, GPU Operator	Same + Harbor registry with local cache + Rancher + Network Operator	The registry cache is a Nepal-specific win: image pulls are the slowest part of every job when they cross the border.
Security	Keycloak, DNS/NTP	Keycloak + Vault + cert-manager + Kyverno + Falco + audit logs + Velero	This is the SOC 2 checklist. v1 has none of it, which pushes the M9 Type I target out by however long it takes to add.
Observability	—	DCGM + Prometheus + Grafana + Loki + OTel + facility telemetry	No monitoring means no usage billing, no SLA proof and no early warning of a dying GPU. It is the layer everything else reports into.
Operating layer	RKE2, CUDA, Ubuntu + Rocky, optional Slurm	RKE2, CUDA via Operator, Ubuntu (POC) / Talos (the future phase), MAAS/Tinkerbell, node health loop, BMC	Same core. v2 adds how a node gets built and how a bad one gets pulled, which is most of the day-to-day ops work.
Infrastructure	Servers / Storage / Networking / Accelerators (blank)	X10000 at POC → Ceph at the future phase; InfiniBand fabric; front-end network; facility telemetry	Storage and fabric are the second and third largest capex lines. Leaving them blank means the TCO model is missing them too.

Pros and cons

Your v1 draft

Lean, all open source, right instincts.

PROS

- The core is correct: Kubernetes-native, GitOps-driven, NVIDIA GPU Operator, hardened RKE2. This is how CoreWeave-class clouds are built.
- Small enough for two or three engineers to hold in their heads. Nothing exotic.
- HAMi is a sound choice for the Shared tier, and it already supports MIG for the POC plan.
- Zero licence cost in the diagram as drawn.
- Slurm is acknowledged, which most GPU-cloud plans forget until a university asks.

CONS

- OpenStack beside Kubernetes is two clouds to operate. For a small team that is the single most expensive line in the diagram.
- No monitoring, so no usage-based billing, no SLA evidence, no GPU health signal.
- Billing, tenancy and catalogue are all hand-written. That is 6–12 months of engineering before the first invoice.
- No job queue, so tiered pricing has no enforcement mechanism.
- Security and audit tooling absent; SOC 2 cannot start.
- Storage, fabric and registry unspecified, so both the architecture and the capex model have holes where the second- and third-largest costs sit.
- Two base OSes, single-user Jupyter, agents with no sandbox.

Preferred v2

Same core, complete picture, less custom code.

PROS

- Every layer a customer, auditor or bank will ask about is drawn and named.
- Billing is derived from measured GPU use end to end, not estimated.
- Custom code shrinks to the portal and the glue. Tenancy, metering, invoicing and the API gateway are assembled, not written.
- Stronger isolation (vCluster + Cilium + Kata) supports the Sovereign tier without separate hardware on day one.
- Fits Option B: Layer 5 runs against HPE during the POC and against RKE2 in the future phase with no customer-visible change.
- Still almost entirely open source; the commercial lines are storage and fabric, which are hardware decisions anyway.

CONS

- Roughly three times as many components. Needs three to four strong platform engineers, not one or two, and a deliberate rollout order.
- Parallel storage and a GPU fabric are real money; they are not in the current TCO floor.
- KubeVirt and Kata add complexity that only pays off if the Dedicated tier and agent hosting actually sell. Defer both until a customer asks.
- Ubuntu at the POC and Talos at the future phase means two node images to maintain; acceptable only because it is one per site.
- OpenMeter and Lago are younger projects than the rest of the stack; keep the billing data model simple so they can be swapped.
- More design work before first revenue, which cuts against "customer acquisition is the constraint". The phasing below is the answer to that.

Net: v1 and v2 agree on the engine. The differences are one removal (OpenStack), one mechanism v1 doesn't have (metering to invoice), one discipline v1 skips (security and audit from day one), and one rule of thumb: build the portal, buy everything behind it.

What to build when

Because the POC runs on HPE's own Kubernetes, most of Layers 2–4 aren't yours to build during the POC. What is yours is the customer-facing layer, and that is the same code either way.

Now · POC on HPE (WorldLink)

Build only what HPE doesn't give you and what migrates unchanged.

- HPE's written answers on running Layer 5 on the management nodes
- Portal, Keycloak realms, vCluster per customer
- DCGM → Prometheus → OpenMeter → Lago pipeline (Layer 3 + 5)
- LiteLLM gateway in front of vLLM (model API product)
- Harbor with pull-through cache
- Audit log shipping, Vault, status page (SOC 2 Type I groundwork)
- Kueue, if HPE's cluster allows it; otherwise design the plans around it for later

Future phase · own data centre

Stand up Layers 1–2 and 4 yourself; move Layer 5 across untouched.

- Talos images + RKE2 + Rancher (MAAS/Tinkerbell postponed)
- GPU Operator, Network Operator, HAMi, Kueue
- Cilium, MetalLB, Envoy Gateway, cert-manager, Kyverno, Falco, Velero
- Ceph (file + object) + InfiniBand fabric (procurement starts during the POC)
- KubeVirt / Kata / Slurm-on-K8s / agent hosting only when a paying customer needs them
- Facility telemetry into Grafana for carbon reporting

Cross-cutting responsibilities

Five things that aren't a single tile but need one head each.

Metering → invoice pipeline

End to end: DCGM → Prometheus → OpenMeter → Lago → Portal. One product, one owner.

Kaustav + Aatiz

SOC 2 Type I programme

Policies, evidence collection, auditor relationship. Owns the checklist in Layer 4.

Aadarsh + Abhigyan

Vendor management

HPE, Dell/Supermicro, storage and fabric vendors, WorldLink. Quotes, terms, escalations.

Abhigyan + Aatiz

Customer onboarding + FDE

Getting each customer's workload running on the platform and keeping them there.

Kaustav + Aatiz

Ops runbooks + on-call

Who gets paged, what they do, and the written procedures behind every alert.

Aatiz + Aadarsh

Decisions

SETTLED GPU sharing: HAMi

Not KAI. First job is proving **MIG mode on the H200s** in the HPE box, since MIG is the POC Shared-tier mechanism.

Aatiz

SETTLED Node OS: Ubuntu + Talos

Ubuntu on the HPE POC, Talos in the future phase. One OS per site; never two in one cluster.

Aatiz

+ Aadarsh

SETTLED GPU fabric: InfiniBand

Applies to the future phase (own data centre). The POC runs on HPE's 400GbE Spectrum fabric as shipped; that does not change.

Aadarsh

+ Aatiz

SETTLED Storage: X10000 now, Ceph later

HPE's Alletra MP X10000 (file + S3) at the POC, untouched. **Ceph on our own nodes in the future phase.** No Ceph on top of HPE storage.

Aatiz

+ Abhigyan

SETTLED Layer 5 runs on the HPE management nodes

In CoreValley namespaces on HPE's Kubernetes (3× DL325). **Pending HPE's written answers:** own charts/operators allowed? survive upgrades? support impact? A separate cluster only if the answer is no.

Abhigyan

+ Aatiz

OPEN Dedicated tier: VMs or whole nodes?

Lean: **whole nodes first.** With 8 GPUs there is little to slice, and buyers want "your server", not "your VM". Add KubeVirt when a named customer needs it.

Abhigyan

+ Aatiz

Who owns what

The same assignments, read by person. **Bold** items are on the POC critical path; the rest are future-phase or HPE-provided until then.

Aatiz CTO · infrastructure & platform · 27 owned · 13 support OWNS HAMi (prove MIG on H200), Kueue, vCluster DCGM exporter, Argo CD (GitOps), Rancher, HPE cluster integration Model API endpoints + LiteLLM gateway, JupyterHub Harbor registry + pull-through cache GPU pods product; Storage fast tier + object - Kubernetes scheduler, GPU Operator, Network Operator, KubeVirt + Kata - Dedicated bare metal; RKE2, Ubuntu + Talos images, CUDA, node health loop, BMC/Redfish - Ops runbooks + on-call POSTPONED Batch/HPC (Slurm-on-K8s), Agent hosting, MAAS/Tinkerbell SUPPORTS - OpenMeter, OpenCost, metering → invoice pipeline, Prometheus, Grafana, Kyverno, Velero, Cilium, Accelerators/Servers spec, GPU fabric, vendor management, onboarding, Dedicated-tier decision	Aadarsh Network & security · 13 owned · 12 support OWNS Vault + External Secrets, cert-manager, audit log shipping Velero backups SOC 2 Type I programme - Kyverno, Falco, Cilium, MetalLB + Envoy Gateway, DNS/NTP - Facility telemetry - GPU fabric (InfiniBand design), front-end network + WAN SUPPORTS - Keycloak, Harbor, Loki + OpenTelemetry, Dedicated tier isolation, Network Operator, KubeVirt + Kata, OS hardening, MAAS/PXE, BMC/OOB network, Facility DCIM, on-call, SOC 2 evidence from every layer	Kaustav Fullstack · FDE · 11 owned · 9 support OWNS Portal (sign-up, catalogue, launch, usage, invoices) Metering → invoice pipeline: OpenMeter, Lago, OpenCost Keycloak (identity, one realm per customer) Prometheus + Alertmanager, Grafana, Loki + OpenTelemetry Status page + support desk Customer onboarding + FDE SUPPORTS - GPU pods, JupyterHub, Model API, LiteLLM gateway, vCluster, Kueue (plan tiers), Argo CD, MetalLB/ingress, DCGM (billing consumer)	Abhigyan MD · vendors, facility, commercial · 8 owned · 6 support OWNS HPE relationship: the three written questions on Layer 5 Vendor management (HPE, Dell/Supermicro, storage, WorldLink) Accelerators + servers procurement Decision: Dedicated tier VMs vs whole nodes - Facility (NEA feeder, UPS, cooling, DCIM) - SOC 2 sponsor SUPPORTS - Portal (product), Lago (plans and prices), status page, storage procurement, front-end/WAN (WorldLink, transit), facility telemetry data
--	--	--	---